

Chapter 2: Atoms, Molecules, and Ions

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PHY201
Lecture 2



Part I

Atomic Theory

Outline of Part I: Atomic Theory

Early Ideas in Atomic Theory

Evolution of Atomic Theory

Outline of Part I: Atomic Theory

Early Ideas in Atomic Theory

Evolution of Atomic Theory

Ancient Atomism

Democritus & Leucippus (~400 BC)

- ▶ Proposed that matter consists of tiny, indivisible particles called **atomos** (Greek: “uncuttable”).
 - ▶ Atoms differ in size, shape, and arrangement.
 - ▶ Empty space separates atoms.
-
- ▶ Aristotle rejected atomism, preferring four “elements”: earth, water, air, fire.
 - ▶ Aristotle’s authority dominated for ~2000 years.

Ancient atomism was **philosophical**, not experimental.

It lacked the evidence needed to be a scientific theory — but it planted the right seed.

atomos [ˈa.tə.məs] *adjective* –
Ancient Greek
indivisible, uncuttable
 α (not) + $\tau\epsilon\mu\nu\omega$ (to cut)

Dalton's Atomic Theory (1807)

Five Postulates

1. Matter consists of extremely small, indivisible particles called **atoms**.
2. All atoms of a given element are **identical** in mass and properties.
3. Atoms of **different elements** have different properties.
4. Compounds contain atoms of two or more elements in **small, whole-number ratios**.
5. Atoms are **neither created nor destroyed** in a chemical reaction; they are rearranged.

Postulate 5 is the atomic-theory basis for the **Law of Conservation of Mass**.

Postulates 2 & 4 explain the **Law of Definite Proportions**.

Postulates 3 & 4 explain the **Law of Multiple Proportions**.

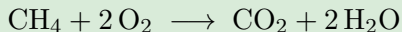
We now know Postulate 2 is not strictly true — **isotopes** exist.

Law of Conservation of Mass

Statement

The total mass of all substances present **does not change** during a chemical reaction.

Example: Combustion of Methane



► Mass of reactants ($\text{CH}_4 + 2 \text{O}_2$) = Mass of products ($\text{CO}_2 + 2 \text{H}_2\text{O}$)

- Atoms are merely **rearranged**, not created or destroyed.
- This law is the reason we balance chemical equations.

Law of Definite Proportions

Statement (Law of Constant Composition)

All pure samples of a given compound contain the same elements in the same **proportions by mass**, regardless of source.

Example: Water

- ▶ Water from a river, a glacier, or a lab all contain 11.19% H and 88.81% O by mass.
- ▶ The ratio is always $\text{H}:\text{O} = 1:8$ by mass.

Mass Fraction

- ▶ **Mass fraction** of element X =
$$\frac{\text{mass of X in sample}}{\text{total mass of sample}}$$
- ▶ Often expressed as a percentage (multiply by 100%).

Law of Multiple Proportions

Statement

When two elements form more than one compound, the masses of one element that combine with a fixed mass of the other are in a **ratio of small whole numbers**.

Example: Two Compounds of Elements B and R — (chem2e_ch2_2.1)

	Compound 1	Compound 2	Ratio of B:R ratios:
			$\frac{1.50}{0.748} \approx \boxed{2.00}$
% B by mass	42.8%	60.0%	$\Rightarrow$ a small whole number!
% R by mass	57.2%	40.0%	$\therefore$ Compound 1 is B R
B:R mass ratio	$42.8/57.2 = 0.748$	$60.0/40.0 = 1.50$	Compound 2 is B R ₂ (or B ₂ R)

Check Your Understanding

Nitrogen and oxygen form several compounds.

In compound A, 14.0 g of N combines with 16.0 g of O.

In compound B, 14.0 g of N combines with 32.0 g of O.

Which law does this illustrate, and what is the ratio?

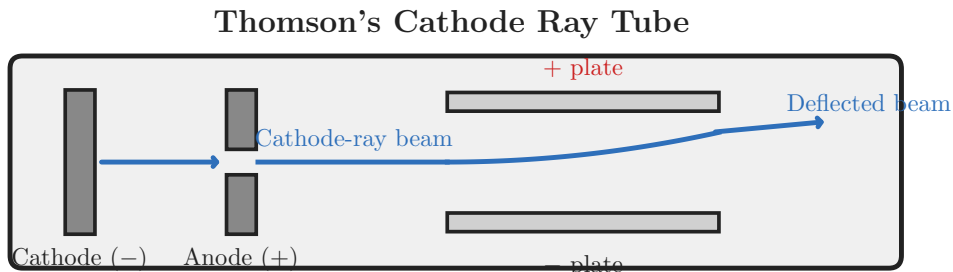
Come to lecture to see the answer.

Outline of Part I: Atomic Theory

Early Ideas in Atomic Theory

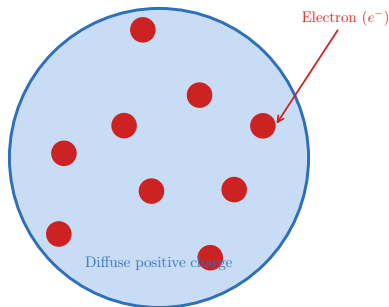
Evolution of Atomic Theory

Discovery of the Electron: Thomson's Cathode Ray Tube



Thomson's Plum Pudding Model

Thomson's Plum Pudding Model

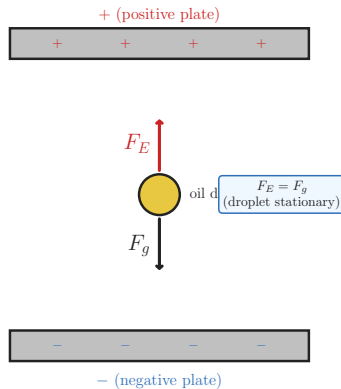


Thomson's Atomic Model

► Atoms are a diffuse sphere of positive charge with electrons embedded

Millikan's Oil Drop Experiment (1909)

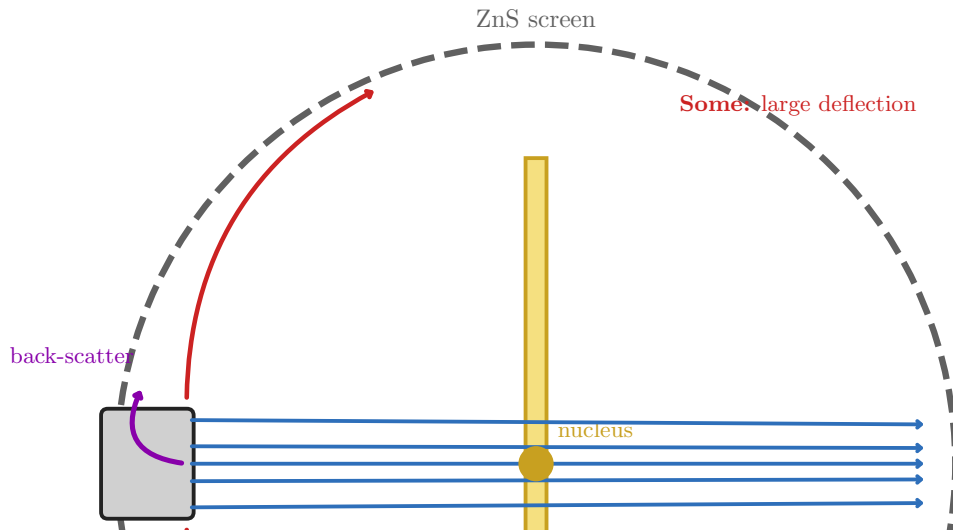
Millikan's Oil Drop Experiment



What Millikan Measured

Rutherford's Gold Foil Experiment (1909)

Rutherford Gold Foil Experiment



Rutherford's Nuclear Model

Rutherford's Conclusion

Atoms contain a **tiny, dense, positively charged nucleus** surrounded by mostly **empty space** through which electrons move.

Key Features

- ▶ Nucleus contains essentially all of the atom's mass.
- ▶ Nucleus diameter: $\sim 10^{-15}$ m ($\sim 10^5$ times smaller than the atom).
- ▶ Electrons occupy the vast empty space outside the nucleus.

Rutherford's famous analogy:

“It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you.”

Later work (Chadwick, 1932) identified **neutrons** — neutral particles in the

Check Your Understanding

In Rutherford's gold foil experiment, what would have happened if Thomson's plum pudding model were correct?

Why was the actual result surprising?

Come to lecture to see the answer.

Part II

Atomic Structure

Outline of Part II: Atomic Structure

Atomic Structure and Symbolism

Outline of Part II: Atomic Structure

Atomic Structure and Symbolism

The Three Subatomic Particles

Particle	Symbol	Charge	Mass (amu)	Location
Proton	p^+	+1	1.00727	Nucleus
Neutron	n^0	0	1.00866	Nucleus
Electron	e^-	-1	0.000549	Outside nucleus

- ▶ 1 amu (atomic mass unit) $\equiv \frac{1}{12}$ the mass of a ^{12}C atom.
- ▶ The electron mass ($\approx 1/1836$ of a proton) is negligible in mass calculations.
- ▶ Protons + Neutrons $\approx$ all the atom's mass.

Atomic Number, Mass Number, and Neutrons

Atomic Number (Z)

Number of **protons** in the nucleus.

Defines the element: every carbon atom has $Z = 6$.

Mass Number (A)

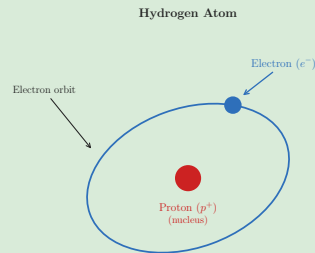
Total number of **protons + neutrons**:

$$A = Z + N$$

where N = number of neutrons.

Neutrons

Hydrogen Atom



- ▶ $Z = 1$ (1 proton)
- ▶ $A = 1$ (1 proton, 0 neutrons)
- ▶ 1 electron (neutral atom)

Nuclide Symbols

Standard Notation

$${}^A_Z\text{X} \quad \text{or equivalently} \quad \text{X-}A$$

where X = chemical symbol, A = mass number, Z = atomic number.

Examples

- ▶ ${}^1_1\text{H}$ (or H-1): 1 proton, 0 neutrons
- ▶ ${}^4_2\text{He}$ (or He-4): 2 protons, 2 neutrons
- ▶ ${}^{12}_6\text{C}$ (or C-12): 6 protons, 6 neutrons
- ▶ ${}^{238}_{92}\text{U}$ (or U-238): 92 protons, 146 neutrons

Z can be omitted in writing because the element symbol already specifies it.

A is always given explicitly because it varies between isotopes.

Isotopes

Definition

Isotopes are atoms of the *same* element (same Z) that differ in the number of neutrons (different A).

Isotopes of Carbon



6 protons

6 neutrons

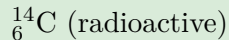
98.89% abundance



6 protons

7 neutrons

1.11% abundance



6 protons

8 neutrons

trace abundance

- ▶ Isotopes are **chemically identical** — same number of electrons, same bonding behavior.
- ▶ Isotopes have different **masses** and may differ in radioactivity.

Ions: Cations and Anions

Ion

An atom (or molecule) with an **unequal number of protons and electrons** — it carries a net charge.

Cation (positive ion)

- ▶ **Loses** one or more electrons.
- ▶ More protons than electrons.
- ▶ *Examples:* Na^+ , Mg^{2+} , Al^{3+}

Anion (negative ion)

- ▶ **Gains** one or more electrons.
- ▶ More electrons than protons.
- ▶ *Examples:* Cl^- , O^{2-} , N^{3-}

Neutral atom: $\# \text{electrons} = \# \text{protons}$

If $\# e^- \neq \# p^+$, we have an ion.

e.g. A sodium atom ($11 p^+$, $11 e^-$) loses one electron $\Rightarrow \text{Na}^+$ ($11 p^+$, $10 e^-$).

Check Your Understanding

An atom of ${}_{17}^{35}\text{Cl}$ gains one electron to form a chloride ion.

How many protons, neutrons, and electrons does the ion have?

What is its charge?

Come to lecture to see the answer.

Average Atomic Mass

Why Average?

Most elements exist as a **mixture of isotopes**. The atomic mass listed on the periodic table is a **weighted average** of all natural isotopes.

Formula

$$\bar{m} = \sum_i (\text{fractional abundance}_i \times m_i)$$

- ▶ Fractional abundance = (percent abundance) / 100
- ▶ m_i = isotopic mass in amu

Isotopic abundances and masses are measured by **mass spectrometry**.

Example: Average Atomic Mass of Boron

Average Atomic Mass of Boron — (chem2e_ch2_ex2.4)

Boron has two stable isotopes:

^{10}B : 19.9% abundance, mass = 10.0129 amu

^{11}B : 80.1% abundance, mass = 11.0093 amu

Calculate the average atomic mass of boron.

$$\bar{m} = (0.199)(10.0129 \text{ amu}) + (0.801)(11.0093 \text{ amu})$$

$$\bar{m} = 1.993 \text{ amu} + 8.818 \text{ amu}$$

$$\bar{m} = 10.811 \text{ amu}$$

The average is **closer to 11** because ^{11}B is more abundant ($\sim 80\%$).

Compare with the periodic table value: 10.81 amu.

Check Your Understanding

Chlorine has two stable isotopes:

^{35}Cl : 75.77% abundance, mass = 34.969 amu

^{37}Cl : 24.23% abundance, mass = 36.966 amu

Without calculating, is the average atomic mass of chlorine closer to 35 or 37?

Come to lecture to see the answer.

Part III

Molecules and the Periodic Table

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

Types of Chemical Formulas

Molecular Formula

Shows the **actual number** of each type of atom in one molecule.

Examples:

H_2O , CO_2 , $\text{C}_6\text{H}_{12}\text{O}_6$

Empirical Formula

Shows the **simplest whole-number ratio** of atoms.

Examples:

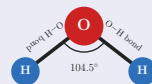
H_2O , CO_2 , CH_2O (for glucose)

Structural Formula

Shows how atoms are **connected** (bonded) to each other.

Structural Formula of Water

Molecular formula: H_2O



Reveals molecular

Subscripts vs. Coefficients

Important Distinction

- ▶ A **subscript** tells how many atoms of that element are in *one* molecule.
- ▶ A **coefficient** tells how many molecules (or formula units) are present.

Examples

- | | |
|--|---|
| ▶ H_2 = one molecule with 2 H atoms bonded together | ▶ H_2O = 2 H + 1 O bonded in one molecule |
| ▶ 2H = two separate H atoms (not bonded) | ▶ $3\text{H}_2\text{O}$ = 3 water molecules (6 H + 3 O total) |
| ▶ 2H_2 = two molecules of H_2 (4 H atoms total) | |

When writing chemical equations, you change **coefficients** to balance — never

Isomers

Definition

Isomers are compounds that share the same **molecular formula** but have **different structural arrangements**, and therefore different properties.

Example: C_2H_6O

Isomers of C_2H_6O

Both: molecular formula C_2H_6O



Molecular Mass (Formula Mass)

Definition

The **molecular mass** (also called formula mass) is the sum of the **atomic masses** of all atoms in one molecule, expressed in amu.

Examples

$$M(\text{H}_2\text{O}) = 2(1.008) + 15.999 = \boxed{18.02 \text{ amu}}$$

$$M(\text{CO}_2) = 12.011 + 2(15.999) = \boxed{44.01 \text{ amu}}$$

$$M(\text{CO}) = 12.011 + 15.999 = \boxed{28.01 \text{ amu}}$$

Note that CO_2 and CO have very different molecular masses despite containing the same elements — because they differ in the **number** of oxygen atoms.

Check Your Understanding

The molecular formula for glucose is $\text{C}_6\text{H}_{12}\text{O}_6$.

- (a) What is its empirical formula?
- (b) How many total atoms are in one glucose molecule?

Come to lecture to see the answer.

Outline of Part III: Molecules and the Periodic Table

Chemical Formulas

The Periodic Table

Organization of the Periodic Table

The Periodic Table of the Elements

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H																	He
2	Li	Be																
3	Na	Mg																
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra	**	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og

*	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
**	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Metals, Nonmetals, and Metalloids

Metals

- ▶ Shiny, malleable, ductile
- ▶ Good conductors of heat and electricity
- ▶ **Lose** electrons to form cations
- ▶ Left and center of periodic table

Nonmetals

- ▶ Dull, brittle (if solid)
- ▶ Poor conductors
- ▶ **Gain** electrons to form anions (or share electrons)
- ▶ Upper right of periodic table

Metalloids

- ▶ Intermediate properties
- ▶ Moderate conductivity (**semiconductors**)
- ▶ *Examples:* Si, Ge, As, Sb, Te
- ▶ Staircase boundary between metals and nonmetals

Transition metals (Groups 3–12) readily form cations of **varying charge**.
Main-group metals typically form cations of a **fixed charge**.

Named Element Families

Group	Name	Key Property	Examples
1	Alkali metals	Very reactive; form 1+ ions	Li, Na, K
2	Alkaline earth metals	Reactive; form 2+ ions	Mg, Ca, Ba
15	Pnictogens	Nitrogen group	N, P, As
16	Chalcogens	Oxygen group; form 2− ions	O, S, Se
17	Halogens	Highly reactive; form 1− ions	F, Cl, Br, I
18	Noble gases	Very unreactive (full shell)	He, Ne, Ar

Elements in the same group have the same number of **valence electrons**, which determines their chemistry.

Check Your Understanding

Without looking at a periodic table, classify each element as a **metal**, **nonmetal**, or **metalloid**, and identify its group family if it has a named one:

- (a) Potassium (K, $Z = 19$)
- (b) Fluorine (F, $Z = 9$)
- (c) Silicon (Si, $Z = 14$)
- (d) Argon (Ar, $Z = 18$)

Come to lecture to see the answer.

Part IV

Compounds and Nomenclature

Outline of Part IV: Compounds and Nomenclature

Ionic and Molecular Compounds

Chemical Nomenclature

Outline of Part IV: Compounds and Nomenclature

Ionic and Molecular Compounds

Chemical Nomenclature

Ion Formation

Why Ions Form

During chemical reactions, **electrons transfer** from one atom to another, producing electrically charged particles (ions).

Metals → Cations

- ▶ Metals **lose** electrons to form cations.
- ▶ Group 1 metals: lose $1\text{ e}^- \Rightarrow 1+$ charge
e.g. $\text{Na} \rightarrow \text{Na}^+$
- ▶ Group 2 metals: lose $2\text{ e}^- \Rightarrow 2+$ charge
e.g. $\text{Mg} \rightarrow \text{Mg}^{2+}$
- ▶ Group 13 metals: lose $3\text{ e}^- \Rightarrow 3+$

Nonmetals → Anions

- ▶ Nonmetals **gain** electrons to form anions.
- ▶ Group 17 (halogens): gain $1\text{ e}^- \Rightarrow 1-$ charge
e.g. $\text{Cl} \rightarrow \text{Cl}^-$
- ▶ Group 16 (chalcogens): gain $2\text{ e}^- \Rightarrow 2-$ charge
e.g. $\text{O} \rightarrow \text{O}^{2-}$
- ▶ Group 15: gain $3\text{ e}^- \Rightarrow 3-$ charge

Common Monatomic Ions

Common Cations

Na^+	sodium	Fe^{2+}	iron(II)
K^+	potassium	Fe^{3+}	iron(III)
Mg^{2+}	magnesium	Cu^+	copper(I)
Ca^{2+}	calcium	Cu^{2+}	copper(II)
Ba^{2+}	barium	Zn^{2+}	zinc
Al^{3+}	aluminum	NH_4^+	ammonium

Common Anions

F^-	fluoride	O^{2-}	oxide
Cl^-	chloride	S^{2-}	sulfide
Br^-	bromide	N^{3-}	nitride
I^-	iodide	P^{3-}	phosphide

Monatomic anion names end in **-ide**.

Polyatomic Ions

Definition

A **polyatomic ion** is a charged group of two or more bonded atoms that acts as a single unit.

Formula	Name	Charge	Formula	Name	Charge
NH_4^+	ammonium	1+	OH^-	hydroxide	1-
NO_3^-	nitrate	1-	NO_2^-	nitrite	1-
CO_3^{2-}	carbonate	2-	HCO_3^-	bicarbonate	1-
SO_4^{2-}	sulfate	2-	SO_3^{2-}	sulfite	2-
PO_4^{3-}	phosphate	3-	CN^-	cyanide	1-

These **must be memorized**. They appear throughout the rest of chemistry.

Ionic Compounds

Definition

An **ionic compound** consists of cations and anions arranged in a crystal lattice, held together by **electrostatic attraction** (ionic bonds).

Writing Ionic Formulas

- ▶ The overall compound must be **electrically neutral**.
- ▶ Cation charge + anion charge = 0.
- ▶ *Example:* Mg^{2+} and F^{-}
Need 2 F^{-} per Mg^{2+} : **MgF_2**
- ▶ *Example:* Mg^{2+} and OH^{-}
Need 2 OH^{-} per Mg^{2+} : **$\text{Mg}(\text{OH})_2$**

Ionic compounds:

- ▶ Typically have **high melting points**.
- ▶ Conduct electricity when **molten** or **dissolved in water**.
- ▶ Often form crystalline solids.

Molecular (Covalent) Compounds

Definition

A **molecular compound** (covalent compound) consists of discrete molecules formed when nonmetals **share electrons** via covalent bonds.

Examples

- ▶ H_2O (water)
- ▶ CO_2 (carbon dioxide)
- ▶ NH_3 (ammonia)
- ▶ CH_4 (methane)
- ▶ N_2O_4 (dinitrogen tetroxide)

Molecular vs. Ionic

Molecular	Ionic
Nonmetal + nonmetal	Metal + nonmetal
Share electrons	Transfer electrons
Discrete molecules	Crystal lattice
Lower melting pt	Higher melting pt

Check Your Understanding

Classify each as **ionic** or **molecular**:

- (a) NaCl (table salt)
- (b) H₂O (water)
- (c) MgO (magnesium oxide)
- (d) SO₂ (sulfur dioxide)

Come to lecture to see the answer.

Outline of Part IV: Compounds and Nomenclature

Ionic and Molecular Compounds

Chemical Nomenclature

Naming Ionic Compounds: Monatomic Ions

Rule

Name the **cation** first, then the **anion** with an **-ide** suffix.

Examples

- ▶ NaCl = sodium chloride
- ▶ KBr = potassium bromide
- ▶ MgO = magnesium oxide
- ▶ CaF_2 = calcium fluoride
- ▶ Al_2O_3 = aluminum oxide
- ▶ MgF_2 = magnesium fluoride

- ▶ Do **not** include subscripts in the name.
- ▶ Do **not** say “magnesium di-fluoride” — the charge ratio is already fixed by the ion charges.

Naming Ionic Compounds: Variable-Charge Metals

Rule

When a metal can form more than one cation charge, indicate the charge with a **Roman numeral** in parentheses after the metal name.

Examples

- ▶ FeCl_2 = **iron(II) chloride** (Fe^{2+})
- ▶ FeCl_3 = **iron(III) chloride** (Fe^{3+})
- ▶ CuO = **copper(II) oxide** (Cu^{2+})
- ▶ Cu_2O = **copper(I) oxide** (Cu^{+})
- ▶ SnCl_4 = **tin(IV) chloride** (Sn^{4+})

Metals with Variable Charge

- ▶ Fe (iron): 2+, 3+
- ▶ Cu (copper): 1+, 2+
- ▶ Sn (tin): 2+, 4+
- ▶ Pb (lead): 2+, 4+
- ▶ Hg (mercury): 1+, 2+
- ▶ Co, Cr, Mn, Ni ...

Naming Ionic Compounds: Polyatomic Ions

Rule

Use the **name of the polyatomic ion** directly; do not add -ide unless the ion is CN^- (cyanide) or OH^- (hydroxide).

Examples

- ▶ NaOH = sodium hydroxide
- ▶ $\text{Al}_2(\text{SO}_4)_3$ = aluminum sulfate
- ▶ $\text{Mg}(\text{NO}_3)_2$ = magnesium nitrate
- ▶ $(\text{NH}_4)_2\text{CO}_3$ = ammonium carbonate
- ▶ $\text{Ca}_3(\text{PO}_4)_2$ = calcium phosphate
- ▶ $\text{Mg}(\text{OH})_2$ = magnesium hydroxide

Note parentheses: use them around a polyatomic ion when the subscript > 1 .

Naming Ionic Hydrates

Definition

A **hydrate** is an ionic compound with a specific number of water molecules incorporated into its crystal structure.

Naming Rule

Name the ionic compound normally, then add a Greek prefix + **hydrate**.

Examples

- ▶ $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ = **copper(II) sulfate pentahydrate**
- ▶ $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ = **calcium chloride dihydrate**
- ▶ $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ = **sodium sulfate decahydrate**

The “ $\cdot$ ” (centered dot) in the formula separates the ionic compound from the

Check Your Understanding

Name each compound:

- (a) K_2S
- (b) Fe_2O_3
- (c) $\text{Mg}(\text{HCO}_3)_2$
- (d) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Come to lecture to see the answer.

Naming Molecular Compounds: Greek Prefixes

Rule

Use **Greek prefixes** to indicate the number of each type of atom. Name the more metallic element first; second element gets an **-ide** suffix. Drop the trailing vowel of the prefix before a vowel (e.g. monooxide → monoxide).

Prefixes

1	mono-	6	hexa-
2	di-	7	hepta-
3	tri-	8	octa-
4	tetra-	9	nona-
5	penta-	10	deca-

Examples

- ▶ CO_2 = carbon dioxide
- ▶ CO = carbon monoxide
- ▶ SO_2 = sulfur dioxide
- ▶ N_2O_4 = dinitrogen tetroxide
- ▶ NO_2 = nitrogen dioxide
- ▶ PCl_5 = phosphorus pentachloride
- ▶ SF_6 = sulfur hexafluoride

Naming Binary Acids

Binary Acid

A **binary acid** consists of hydrogen bonded to one nonmetal element, dissolved in water.

Naming Rule

1. Replace “hydrogen” with the prefix **hydro-**.
2. Change the nonmetal ending to **-ic**.
3. Add the word **acid**.

Examples

- ▶ HCl (pure gas): hydrogen chloride
HCl(aq): **hydrochloric acid**
- ▶ HBr(aq): **hydrobromic acid**
- ▶ HF(aq): **hydrofluoric acid**

Naming Oxyacids

Oxyacid

An **oxyacid** contains hydrogen, oxygen, and at least one other element.

Naming Rule

Start from the name of the **oxyanion**, then:

- ▶ If anion ends in **-ate** $\Rightarrow$ acid ends in **-ic acid**
- ▶ If anion ends in **-ite** $\Rightarrow$ acid ends in **-ous acid**

Examples

- | | |
|--|---|
| ▶ NO_3^- = nitrate $\Rightarrow$ HNO_3 = nitric acid | ▶ SO_3^{2-} = sulfite $\Rightarrow$ H_2SO_3 = sulfurous acid |
| ▶ NO_2^- = nitrite $\Rightarrow$ HNO_2 = nitrous acid | ▶ CO_3^{2-} = carbonate $\Rightarrow$ H_2CO_3 = carbonic acid |

Check Your Understanding

Give the name or formula as appropriate:

- (a) N_2O_5 (molecular compound)
- (b) **sulfur tetrafluoride**
- (c) H_2SO_4
- (d) **hydroiodic acid**

Come to lecture to see the answer.

Check Your Understanding

Write the formula for each compound:

- (a) calcium phosphate
- (b) iron(III) nitrate
- (c) ammonium sulfate
- (d) dinitrogen monoxide

Come to lecture to see the answer.

Summary: Chapter 2

- ▶ **Dalton's atomic theory** (1807): atoms are small, indivisible, and rearranged (not destroyed) in reactions. Explains Laws of Conservation of Mass, Definite Proportions, and Multiple Proportions.
- ▶ **Subatomic particles**: protons (+, in nucleus), neutrons (0, in nucleus), electrons (−, outside nucleus).
- ▶ **Atomic number** $Z = \#$ protons; **mass number** $A =$ protons + neutrons; **isotopes** have same Z , different A .
- ▶ **Average atomic mass** = weighted average over natural isotopes.
- ▶ **Molecular formula** = actual atom count; **empirical formula** = simplest ratio.
- ▶ The **periodic table** organizes elements by increasing Z into periods (rows) and groups (columns) of similar properties.
- ▶ **Ionic compounds**: cation + anion; electrically neutral crystal lattice.
- ▶ **Molecular compounds**: nonmetals sharing electrons.